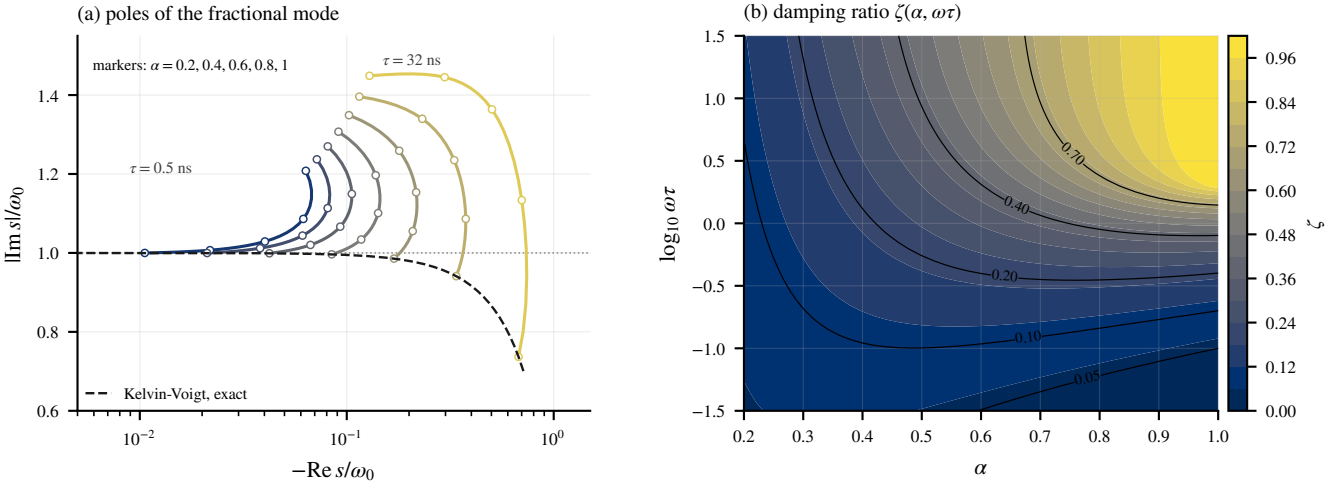
(a) poles of the fractional mode
markers: $\alpha = 0.2, 0.4, 0.6, 0.8, 1$
$\tau = 32$ ns
$\tau = 0.5$ ns
Kelvin-Voigt, exact
$\mathrm{Im}\, s/\omega_0$
$-\mathrm{Re}\, s/\omega_0$
1.4
1.2
1.0
0.8
0.6
10^{-2}
10^{-1}
10^{0}
(b) damping ratio $\zeta(\alpha, \omega\tau)$
$\log_{10} \omega\tau$
α
0.70
0.40
0.20
0.10
0.05
1.5
1.0
0.5
0.0
−0.5
−1.0
−1.5
0.2
0.3
0.4
0.5
0.6
0.7
0.8
0.9
1.0
ζ
0.96
0.84
0.72
0.60
0.48
0.36
0.24
0.12
0.00